MILITARY SPECIFICATION

CABINET, PHOTOGRAPHIC STORAGE, FILM BM-11( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of cabinet having a capacity for storing nine hundred 100-foot reels of 16mm microfilm and designated Cabinet, Photographic Storage, Film BM-11( ) (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

AA-F-791	- Furniture And Cabinets, Office, Sectional, Steel.

MILITARY

MIL-P-116	- Preservation, Methods Of.
MIL-P-11268	- Parts, Materials, And Processes Used In Electronic Equipment.
MIL-F-14072	- Finishes For Ground Electronic Equipment.
MIL-E-55585	- Electronics Equipment And Parts, Packaging Of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Electronics Research and Development Command, Combat Surveillance and Target Acquisition Laboratory, ATTN: DELCS-PE, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards For Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-130	- Identification Marking Of US Military Property.
MIL-STD-726	- Packaging Requirements Code.
MIL-STD-810	- Environmental Test Methods.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E 290-80	- Semi-Guided Bend Test For Ductility Of Metallic Materials.

(Application for copies should be addressed to the American Society For Testing And Materials, 1916 Race Street, Philadelphia, PA 19103.

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The cabinet shall be sheet steel with 9 drawers, lock, drawer latches, and drawer indices as specified herein.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.3 Parts, materials, and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirement therein, shall apply (see 4.4).

3.3.1 Sheet Steel. Sheet steel used in the construction of exposed surfaces, such as drawer fronts, tops, sides and backs shall be of either full pickled, full cold-rolled, reannealed, stretcher leveled, and oiled steel or hot-rolled steel with one pass of cold rolling, and then annealed and pickled. For unexposed parts such as drawer bodies, bottoms, and small parts, the sheet steel shall be the same as for exposed parts with the exception that it may be single pickled and the stretcher leveling may be omitted. The sheet steel shall be bright, smooth and free of rust, scale, pits, scratches, seams, laps, snakes, inclusions, crimps, buckles, and shall be uniform gage; commercial variations being permitted along or across any one sheet. The sheet metal shall be formed into the required shapes without fracture or near fracture, and shall meet the test requirements of ASTM E 290-80.

3.3.2 Recycled, virgin and reclaimed materials. It is required that, to the maximum extent possible, recovered or reclaimed materials, in lieu of virgin materials, shall be used without jeopardizing the intended use of the manufactured item.

3.4 Screws, nuts and other threaded parts. Screw threads shall conform to FED-STD-H28 and shall be class 2 fit unless otherwise specified herein. Threads in soft materials shall be unified course-thread series; threads in other material shall be unified course-thread or fine-thread series, as suitable for the particular application.

3.4.1 Length and engagement of threads. Thread engagement shall be not less than 3 complete threads. When permitted by design of the equipment, screws and bolts shall extend not less than 1.5 threads beyond the nuts or equivalent engaging parts.

3.4.2 Flat head and oval head screws. Counter-sinking shall provide complete seating of flat and oval head screws. Such screws shall not be used to hold material whose thickness is less than 1.5 times the height of the taper on the screw head unless the material is suitably embossed (cupped) to fit the screw head.

3.5 Construction. Construction, in general, shall conform to the applicable portions of AA-F-791. The cabinet shall be steel conforming to 3.3.1 and of the thickness specified in table I. Commercial tolerances are acceptable, except as otherwise specified herein.

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TABLE I. Steel thickness.

Section	Thickness (inch)	Thickness (mm)	Section	Thickness (inch)	Thickness (mm)
Back	0.0375	0.95	Side	0.0375	0.95
Bottom	.050	1.27	Top	.050	1.27
Drawer back	1/		Vertical frame center	.050	1.27
Drawer body	.0250	0.64	Vertical frame center	.062	1.57
Drawer front	.0375	0.95	Vertical frame rear	.062	1.57
Inner front	.0250	0.64			

1/ Straight back shall be not less than 0.050 inch (1.27mm) thick. Folded, beaded, or reinforced with wire across top backs shall be not less than 0.0312 inch (0.79mm) thick.

3.5.1 Structural strength. The cabinet shall be sufficiently rigid so that movable parts, such as drawers, shall not bind or sag to an extent that sagging interferes with ease of operation when the cabinet is supported on diagonally opposite corners and loaded with 25 pounds (11.34kg) weight for each cubic foot (0.028 m^3) of filing space (evenly distributed between or among filing drawers) plus a top load equal to the cabinet weight plus 225 pounds (102.06kg).

3.5.2 Base. The cabinet shall possess a fully closed base.

3.5.3 Lock. Each cabinet shall be fitted with a general lock to prevent opening of any and all drawers. The locks shall permit closing of drawers after the locking mechanism has been placed in the locked position. The locks shall be key type unless a combination type lock is specified (see 6.2). Key type locks shall have not less than four tumblers. Combination type locks shall possess not less than three tumblers. Combination type locks shall be capable of being reset by hand or key.

3.5.4 Drawers. The cabinet shall be provided with 9 drawers and each drawer shall have a capacity of 100 each 100-foot rolls of 16mm microfilm. Each drawer shall have 4 compartments and each compartment shall have a capacity of 25 each of the above specified film rolls. Each drawer shall be provided with a progressive type, suspension cradle equipped with sufficient number of rollers and bearings to permit easy opening and closing. The suspension cradle shall be capable of holding a fully loaded drawer parallel with the cabinet case.

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3.5.4.1 Latches. All drawers shall be fitted with latches capable of holding the drawers securely in place and being readily released.

3.5.4.2 Drawer indices. Each drawer shall be provided with a metallic holder (plate) for holding drawer index or content card. Each holder shall be provided with a translucent card cover.

3.6 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable. Where dimendions are not specified, contractor design limits shall determine compliance with dimensional requirements.

3.7 Environmental.

3.7.1 Humidity. When subjected to the test specified in 4.6.5, the cabinet shall not bind or sag to the extent that sagging interferes with moveable parts.

3.8 Dimensions. The overall dimensions of the cabinet shall be not more than 52 inches (1.32m) high, 20 inches (50.8cm) wide and 28 inches (71.1cm) deep.

3.9 Weight. The weight of the empty cabinet shall be not more than 250 pounds (113.4kg).

3.10 Finish, protective. The cabinet shall be given protective finish in accordance with MIL-F-14072. This includes finishes of hardware such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be final film E, semi-gloss, light gray enamel, conforming to MIL-F-14072.

3.11 Marking. Identification marking shall conform to MIL-STD-130.

3.12 Workmanship. The cabinet shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs herein:

3.4, 3.4.1, 3.4.2	Securing of parts
3.5, 3.8	Dimensions
3.5.4	Drawer capacity and operation
3.5.4.1	Latches
3.5.4.2	Index holders
3.9	Weight

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection (see 4.5).

d. Packaging inspection (see 4.7).

4.3 First article.

4.3.1 First article units. When required, and as specified in the contract, the contractor shall furnish first article units of the complete cabinet.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for group A, group B and group C (see table II, 4.5.3 and 4.5.4). The inspection shall be performed in the following order: 4.4, group A and group B for all units; and group C as specified in 4.5.4. After completion of group C environmental tests, conforming units shall be reinspected and shall pass all group A inspection. Failure of any inspection shall be cause for rejection.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspections required by this specification:

Item	Where required
Materials	3.3.1
Construction	3.5
Finish	3.10

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4.5 Quality conformance inspection.

4.5.1 Inspection. Quality conformance inspection shall be as specified in 4.4 and 4.5.1 through 4.5.4.

4.5.2 Group A inspection. Each unit shall be inspected for conformance to the inspections specified in table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined, or specified herein, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQLs indicated in table II.

TABLE II . Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical	3.12	4.6.1	1.0	4.0
Operation	3.5.1, 3.5.4	4.6.4	1.0	1/

1/ Operational defects are all in major category.

4.5.2.1 Order of inspection within group A. Group A inspection shall be performed in any order that is satisfactory to the Government.

4.5.3 Group B inspection. Group B inspection shall be performed in inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall be as specified in 4.6.3.

4.5.3.1 Group B sampling plan. Sampling shall be in accordance with MIL-STD-105, inspection level S-4. AQL shall be 6.5 percent defective.

4.5.4 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspection. The inspection shall consist of the inspection specified in 4.6.5. Samples shall be selected in accordance with 4.5.4.1.

4.5.4.1 Group C sampling plan. Two sample units shall be selected at random from every 50 or a fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

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4.5.4.2 Group C failures. Action required relative to group C failures shall be as specified (see 6.2).

4.5.4.3 Reinspection of conforming group B and C sample units. Unless otherwise specified, sample units which have been subject to and passed both group B and C inspection may be accepted on the contract provided all damage is repaired and the sample units ae resubjected to and pass group A inspection.

4.6 Methods of inspection.

4.6.1 Visual and mechanical inspection. Cabinet(s) shall be inspected for conformance to 3.12. Major and minor defects shall be classified according to the MIL-STD-105 defintion of major and minor defects.

4.6.2 Marking. Marking shall be inspected for conformance to MIL-STD-130. Missing data shall be a major defect. Illegible marking which can be reworked shall be a minor defect.

4.6.3 Inspection for dimensional interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cabinet drawers and insertable assemblies, such as telescoping sections, when such dimensions affect mating of parts.

b. Location of hinges and fasteners on separable parts or assemblies which must mate.

c. Location of connectors, locking devices, fasteners, and mountings which receive mating parts.

d. Size and form of special threads.

4.6.4 Operational test. Examination and tests of the cabinet, with the drawers and top loaded as specified in 3.5.1 shall be made to determine ease of operation of drawers as specified in 3.5.4. Operation of locks, latches and other moveable parts shall be checked for ease of operation. Loading shall be in place for 24 hours prior to beginning of tests. Nonconformance to 3.5.1 and 3.5.4, or locks, latches and other moveable parts that are inoperable or binding shall constitute failure of this test.

4.6.5 Humidity. The cabinet shall be subjected to the humidity test specified in MIL-STD-810, method 507.1, procedure II. Operation is not required during the humidity test. After the cabinet has been removed from the test chamber it shall be tested as specified in 4.6.4. Nonconformance to 3.7.1 shall constitute failure of this test.

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4.7 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First article inspection of packaging.

b. Quality conformance inspection of packaging.

4.7.1 First article inspection of packaging. Unless otherwise specified in the contract, first article inspection of packaging shall be in accordance with the unit pack design validation requirements of MIL-P-116.

4.7.2 Quality conformance inspection of packaging.

4.7.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.7.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B of quality conformance inspection requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.7.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table III, "Packing inspection provisions." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and acceptance quality level of 4.0 percent defective.

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TABLE III. Packing inspection provisions.

No.	Characteristic	Method of inspection
101	Intermediate container not as specified.	Visual
102	Improper closure of intermediate container.	Visual
103	Shipping containers not in accordance with specification.	Visual
104	Excessive cube.	Visual
105	Improper blocking and bracing.	Visual
106	Closure not in accordance with specification.	Visual
107	Weight and size exceed container limitations.	Weight and measure
108	Strapping not in accordance with specification, incorrectly applied, or omitted.	Visual
109	Marking omitted, incorrect, or illegible.	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. 1Ø-1-1-ØØ-OO-NS-X-OR-Ø-OO-A.

5.1.2 Level B. 1Ø-1-1-ØØ-OO-NS-X-OR-Ø-OO-B.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. Cabinet, Photographic Storage, Film BM-11( ) is a nine-drawer unit designed for the storage of processed 16mm microfilm.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. When a first article is required for inspection and approval (see 3.2, 4.3, and 6.3).

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c. When a combination lock is required (see 3.5.3).

d. Group C failure action (see 4.5.4.2).

e. Level A or B preservation and packing (see section 5).

f. When first article inspection rough handling tests are not required.

g. When first article packaging inspection test reports require acquisition activity approval prior to production unit packing.

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of units as specified in the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.4 Nomenclature. The parentheses in the nomenclature will be replaced by a letter identifying the particular design; for example, BM-11W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Level B preservation. When level B preservation is required, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - AS

Preparing activity:
Army - ER

(Project 6760-0574)

Review activity:
DLA - GS

User activity:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-10147D	CABINET, PHOTOGRAPHIC STORAGE, FILM BM-11( )

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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